Learning Objective 3.4: Calculate the phase weights required to steer a beam to a given angle and predict the resulting array pattern.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **From geometry to phase.** The course array is a uniform linear array of $N = 8$ patch elements spaced $d = 14$ mm apart, receiving at $f = 10.3$ GHz. A plane wave arrives $\theta_0 = 25°$ from broadside.

   (a) In one or two sentences, explain which element the wavefront reaches first and why a fixed phase applied to each element can make all eight received signals add together.

   The wave arrives from the $+25°$ side, so it reaches the highest-numbered element first and each lower-numbered element a little later. Successive elements therefore see the same signal with a fixed phase difference between neighbors. Adding a compensating phase to each element cancels that difference, so all eight signals arrive at the summing junction in phase and add coherently.

   (b) Compute the wavelength, the electrical spacing d/λ, and the product kd in degrees.

   $kd = \boxed{173.2°}$

   $$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{10.3 \times 10^9} = 0.0291 \text{ m} = 29.1 \text{ mm}$$
   $$\frac{d}{\lambda} = \frac{14}{29.1} = 0.481$$
   $$kd = 2\pi\,(0.481) = 3.02 \text{ rad} = 173.2°$$

   (c) Compute the progressive element-to-element phase $\Delta\phi$ required to steer the beam to 25°.

   $\Delta\phi = \boxed{73.2°}$

   $$\Delta\phi = kd\sin(\theta_0) = 173.2° \times \sin(25°)$$
   $$= 173.2° \times 0.4226 = 73.2°$$

   (d) Compute the arrival-time difference between neighboring elements at this steer angle, in picoseconds.

   $\Delta t = \boxed{19.7 \text{ ps}}$

   $$\Delta t = \frac{d\sin(\theta_0)}{c} = \frac{(0.014)\,(0.4226)}{3 \times 10^8}$$
   $$= 1.97 \times 10^{-11} \text{ s} = 19.7 \text{ ps}$$

2. **The phase table.** The same array is commanded to $\theta_0 = -40°$. Element 0 is the phase reference, and the ADAR1000 accepts a phase setting between 0° and 360°.

   (a) Compute $\Delta\phi$ for this steer angle, including its sign.

$$\begin{aligned}\Delta\phi &= 173.2° \times \sin(-40°) \\ &= 173.2° \times (-0.6428) = -111.3°\end{aligned}$$

$\Delta\phi = \boxed{-111.3°}$

   (b) Compute the commanded phase of element 5 from $\phi_n = -n\Delta\phi$, before any wrapping.

$\phi_5 = \boxed{556.6°}$

$$\phi_5 = -5\,(-111.3°) = +556.6°$$

   (c) Give the value actually written to element 5, wrapped into 0° to 360°.

$\phi_5 = \boxed{196.6°}$

$$\phi_5 = 556.6° - 360° = 196.6°$$

   One whole turn is removed.

   (d) Wrapping changed the number by 360° and did not change the pattern. Explain why in one or two sentences, and state what information the wrapped table no longer carries.

   A phase difference of 360° is one full cycle of the carrier, so the element radiates exactly the same field either way; at a single frequency the two settings are indistinguishable. What is lost is the record of the true time delay: the unwrapped ramp corresponds to a specific delay in seconds, while the wrapped values only reproduce that delay at the design frequency. That is why a wideband array needs time-delay units rather than phase shifters alone.

3. **The steered pattern.** The array is steered to $\theta_0 = 50^\circ$. Use $\psi = kd\,(\sin(\theta) - \sin(\theta_0))$ with $kd = 173.2^\circ$, and recall $\lambda/Nd = 29.1/112 = 0.260$.

   (a) Compute the array-factor argument ψ at the observation angle $\theta = 30^\circ$.

   $\psi = \boxed{-46.1^\circ}$

   $$\begin{aligned}\psi &= 173.2^\circ\,(\sin(30^\circ) - \sin(50^\circ)) \\ &= 173.2^\circ\,(0.500 - 0.766) = -46.1^\circ\end{aligned}$$

   (b) Locate the first null on the broadside side of the main lobe from $\sin(\theta) = \sin(\theta_0) - \lambda/Nd$.

   $\theta = \boxed{30.4^\circ}$

   $$\begin{aligned}\sin(\theta) &= 0.766 - 0.260 = 0.506 \\ \theta &= \arcsin(0.506) = 30.4^\circ\end{aligned}$$

   The main lobe therefore reaches 19.6° below its peak.

   (c) The matching null on the other side requires $\sin(\theta) = 0.766 + 0.260 = 1.026$. Explain what that result means physically and what the pattern looks like on that side of the beam.

   No real angle has a sine greater than one, so that null lies outside visible space and never appears in the pattern. The main lobe runs from its peak out toward endfire without closing into a null, so the upper side of the beam is a shoulder that stays well above the sidelobe floor rather than a symmetric lobe edge. The beam is asymmetric about 50°.

   (d) Two commanded angles differ by 5° near broadside, and two others differ by 5° near 60°. Which pair requires the larger change in $\Delta\phi$, and what does that mean for an array searching a sector in equal phase steps?

   The pattern is rigid in $\sin(\theta)$, and $\Delta\phi$ is proportional to $\sin(\theta_0)$, so a 5° move near broadside changes $\sin(\theta_0)$ much more than the same move near 60°: the pair near broadside needs the larger phase change. An array stepped in equal increments of $\Delta\phi$ therefore takes small angular steps near broadside and large ones near endfire, so a uniform search grid in angle requires unequal phase steps, and the beam positions crowd together near broadside if you step the phase uniformly instead.

4. **Scanning costs beamwidth.** Continue with $N = 8$, $d = 14$ mm, $\lambda = 29.1$ mm. Use $\theta_{\text{HP}} \approx 0.886\lambda / (Nd\cos(\theta_0))$ and the broadside uniform-array directivity $D \approx 2Nd/\lambda$.

   (a) Compute the broadside half-power beamwidth of the array.

   $\theta_{\text{HP}} =$ $\boxed{13.2^\circ}$

   $$\theta_{\text{HP}}(0) = \frac{0.886(29.1)}{112} = 0.2302 \text{ rad}$$
   $$= 13.2^\circ$$

   This is the canonical broadside beamwidth of the course array.

   (b) Compute the half-power beamwidth when the beam is steered to $\theta_0 = 55^\circ$.

   $\theta_{\text{HP}} =$ $\boxed{23.0^\circ}$

   $$\theta_{\text{HP}}(55^\circ) = \frac{13.2^\circ}{\cos(55^\circ)} = \frac{13.2^\circ}{0.574}$$
   $$= 23.0^\circ$$

   (c) Compute the broadside directivity of the array in dBi . Then, using the fact that the effective aperture shrinks to its projection $Nd\cos(\theta_0)$ when the beam is steered, compute the expected gain reduction at $\theta_0 = 55^\circ$ and name the effect.

   $D(0) =$ $\boxed{8.9 \text{ dBi}}$

   $\Delta G =$ $\boxed{-2.4 \text{ dB}}$

   $$D(0) = \frac{2(112)}{29.1} = 7.70 \rightarrow 10\log_{10}(7.70) = 8.9 \text{ dBi}$$
   $$\Delta G = 10\log_{10}(\cos(55^\circ))$$
   $$= 10\log_{10}(0.574) = -2.4 \text{ dB}$$

   This is **scan loss**. It is carried by the element pattern, not by the array factor: the array factor of an isotropic-element array keeps essentially the same directivity as it scans, and the projected-aperture loss shows up in the pattern only when the element factor is included, which is the subject of Lesson 22.

   (d) A tracking requirement allows the beam to broaden to at most 20°. Find the largest usable scan angle, and state the trade-off this puts on the system design in one sentence.

   $\theta_0 =$ $\boxed{48.7^\circ}$

   $$\cos(\theta_0) \geqslant \frac{13.2^\circ}{20^\circ} = 0.66$$
   $$\theta_0 \leqslant \arccos(0.66) = 48.7^\circ$$

   Field of view is bought with aperture: covering a wider sector at this beamwidth requires more elements, since only a longer array can afford the $\cos(\theta_0)$ loss in projected length.

5. **The inverse problem.** You read the following phase settings out of the beamformer. The array is the course array, $kd = 173.2°$.

n	0	1	2	3	4	5	6	7
ϕ_n (deg)	0	260.7	161.4	62.1	322.8	223.5	124.2	24.9

(a) Difference neighboring elements, unwrap, and report the common step and the value of $\Delta\phi$.

$\Delta\phi = \boxed{+99.3°}$

$$\phi_2 - \phi_1 = 161.4° - 260.7° = -99.3°$$
$$\phi_1 - \phi_0 = 260.7° \rightarrow 260.7° - 360° = -99.3°$$

All seven differences reduce to $-99.3°$, and the ramp is $\phi_n = -n\Delta\phi$, so $\Delta\phi = +99.3°$.

(b) Recover the commanded steer angle.

$\theta_0 = \boxed{+35.0°}$

$$\sin(\theta_0) = \frac{\Delta\phi}{kd} = \frac{99.3°}{173.2°} = 0.573$$
$$\theta_0 = \arcsin(0.573) = 35.0°$$

(c) Suppose the seven differences had come out as $-99.3°$ for six of them and $-84°$ for one. State two things that could produce that result, and what you would check first.

The array is not carrying a single uniform steering ramp. Either one element has a calibration or cabling offset added to its steering phase, or that channel's phase was written incorrectly, for example a wrapped value entered without its whole turn. A taper affects amplitude and not phase, so it is not a candidate. Check the per-element calibration offsets first, then re-read the register value for that one element and compare it with the commanded value.

(d) The ADAR1000 sets phase on a 2.8125° grid. Give the value element 1 actually takes and the resulting error.

$\phi_1 = \boxed{261.6°}$

$$\frac{260.7°}{2.8125°} = 92.69 \rightarrow 93 \text{ steps}$$
$$\phi_1 = 93\,(2.8125°) = 261.6°$$

The error is $+0.9°$, which is within half a step as expected. L26 works out what a grid of this size costs in sidelobe level and null depth.

Documentation: ______________________________